

Homework (Carolina Reaper)

- Q1.** A sound wave's frequency is modeled by $f(x) = (x-3)^2 + 2$. What is the vertex form of this function?
- (A) $f(x) = (x-2)^2 + 1$
 (B) $f(x) = (x-3)^2 + 2$
 (C) $f(x) = (x-4)^2 + 3$
 (D) $f(x) = (x-1)^2 + 4$
 (E) $f(x) = (x+2)^2 - 1$
- Q2.** A painter uses a 16×20 canvas. A quadratic function $f(x) = ax^2 + bx + c$ models the brush strokes on the canvas, where $a = -0.5$, $b = 3$, and $c = -2$. What is the intercept form of $f(x)$?
- (A) $f(x) = (x+1)(x-4)$
 (B) $f(x) = (x-2)(x+1)$
 (C) $f(x) = (x+2)(x-3)$
 (D) $f(x) = (x-1)(x+2)$
 (E) $f(x) = (x+3)(x-2)$
- Q3.** A rhythm pattern is created using the function $f(x) = (x-2)^2 - 3$. What is the vertex of this parabola?
- (A) $(1, -2)$
 (B) $(2, -3)$
 (C) $(3, -1)$
 (D) $(4, 0)$
 (E) $(0, -1)$
- Q4.** The function $f(x) = (x+2)^2 + 1$ models the sound waves of a guitar string. What is the y -intercept of this function?
- (A) -3
 (B) -2
 (C) -1
 (D) 1
 (E) 5
- Q5.** A quadratic function $f(x) = x^2 + 4x + 3$ is given. What is the vertex form of $f(x)$?
- (A) $f(x) = (x+1)^2 - 2$
 (B) $f(x) = (x+2)^2 - 1$
 (C) $f(x) = (x+3)^2 + 2$
 (D) $f(x) = (x+4)^2 + 3$
 (E) $f(x) = (x-1)^2 + 1$
- Q6.** A music composition uses the function $f(x) = x^2 - 5x + 6$. What is the x -intercept of this parabola?
- (A) $x = -2, x = 3$
 (B) $x = 1, x = 4$
 (C) $x = 2, x = 5$
 (D) $x = 3, x = 6$
 (E) $x = 4, x = 7$
- Q7.** The function $f(x) = x^2 + 2x - 6$ models the frequency of a piano string. What is the vertex of this parabola?
- (A) $(-2, -10)$
 (B) $(-1, -7)$
 (C) $(-1, -6)$
 (D) $(-3, -5)$
 (E) $(-2, -5)$
- Q8.** A painter uses a 20×20 canvas. The function $f(x) = x^2 - 4x + 3$ models the brush strokes on the canvas. What is the y -intercept of this function?
- (A) 3
 (B) 2
 (C) 1
 (D) 0
 (E) -1

Open Ended Questions (FRQ)

- Q9.** The function $f(x) = (x-1)^2 - 2$ models the sound waves of a violin string. Find the x -intercepts and vertex of the parabola. Show all work and explain your reasoning.
- Q10.** A quadratic function $f(x) = x^2 + 3x + 2$ is given. Rewrite the function in vertex form and intercept form. Explain the differences between the two forms and provide examples of when to use each.

Chef's Tips

- Tip 1: Review the learning objectives and ensure students understand the concepts of vertex and intercept forms.
- Tip 2: Encourage students to use graphing calculators or software to visualize the parabolas and check their work.
- Tip 3: Provide feedback and guidance on how to improve problem-solving skills and mathematical reasoning.